Structure of White Dwarfs

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This paper aims to investigate the structure of white dwarf stars by utilising equations of hydrostatic equilibrium and equations of state for non-relativistic degenerate electron gas. The main purpose is to use Runge-Kutta integration methods to explore and compare mass-radius relations across different stellar compositions and central densities. The paper focuses primarily on Carbon (C_6^{12}) composition white dwarfs and Iron (Fe_{26}^{56}) composition white dwarfs, with the intent of comparing the effect of the electron fraction on the structure of the white dwarf. The computed mass-radius curves also provide an approximation for the Chandrasekhar limit for each of the compositions.

I. INTRODUCTION

Stars often undergo an evolutionary process determined by their initial masses at formation. Lower and intermediate mass stars evolve through nuclear fusion of hydrogen and helium across the core and outer shells until they are exhausted. Once fusion processes halt, outward radiation pressure drops and is not able to form an adaptive equilibrium with gravity, resulting in the star collapsing under the gravitational forces due to its mass. This collapse leads to an expulsion of outer layers, forming a planetary nebula alongside a white dwarf.

White dwarfs are stellar remnants that are formed due to an equilibrium reached between gravitational forces that compress a star inwards and an outward electron degeneracy pressure. The stellar remnants vary in composition; some contain iron (Fe_{26}^{56}) nuclei and their respective electrons, while others form a Carbon (C_6^{12}) composition as fusion halted during the process of carbon or oxygen formation. The electron degeneracy pressure that resists the gravitational collapse is due to Pauli's Exclusion Principle, which states that fermions (spin-½ particles) cannot occupy identical quantum states (due to the antisymmetry requirement of wavefunctions of identical fermions). We will assume a spherically symmetric and non-rotating star. By this definition, we can state that the hydrostatic equilibrium is only dependent on the gravitational collapse and electron degeneracy pressure.

There is, however, a critical threshold beyond which electron degeneracy pressure cannot maintain in hydrostatic equilibrium against gravitational forces. This principle of understanding electron degeneracy pressure in white dwarfs was strongly contributed towards by S. Chandrasekhar, who went on to define the critical mass of white dwarfs (relative to their composition) known as the Chandrasekhar Limit. The Chandrasekhar limit is heavily dependent on the composition of the white dwarf as well as the relation between a white dwarf's mass and radius. As we approach the Chandrasekhar Limit, electrons are forced by gravitational collapse into higher energy states and extreme relativistic velocities, in which they resist the gravitational collapse further. The internal pressure at this limit is at its maximum. Masses higher than the Chandrasekhar limit for stellar remnants result in a collapse beyond which the electron degeneracy pressure can no longer overcome gravitational forces (which leads to further variants of stellar remnants such as neutron stars or black holes). [1]

Using the theoretical foundations we have observed so far, the goal of this project is to numerically model the structure of white dwarf stars by solving the equations for hydrostatic equilibrium. We then use computational integration techniques (specifically, a Runge-Kutta 4th order integration method) to explore mass-radius relations for white dwarfs with Carbon and Iron compositions, comparing them to known white dwarfs with estimates for their mass and radius. We then determine the Chandrasekhar Limit for the white dwarfs based on the composition by increasing central density to a maximum.

II. WHITE DWARF MODEL AND EQUATIONS OF STATE

To model the structure of the white dwarf, we will base our reasoning on two basic assumptions. The first is to assume that the white dwarf is spherically symmetric to ensure we care further for the general shape and volume of the star (to better determine the mass of the modelled white dwarf) and not as much about the surface, which is prone to constant minor changes. The second assumption is that the white dwarf is strictly non-rotating. This is because adding rotation to white dwarfs actively leads to deformation effects, leading to a star that is no longer spherically symmetric, which results in the requirement

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for more complex mass distributions and an adjustment in the equations of state for white dwarfs.

The condition for hydrostatic equilibrium requires specifically that the gravitational force acting on the fluid elements is balanced by the electron degeneracy pressure gradient. This is expressed mathematically by:

$$\frac{\mathrm{d}P}{\mathrm{d}r} = -\frac{Gm(r)}{r^2}\rho(r) \tag{1}$$

Where $P(r)$ is the electron degeneracy pressure, $\rho(r)$ is the central density of the star as a function of its radius, and $m(r)$ is the mass enclosed by a sphere of radius r. The distribution of the mass in the star can be described using:

$$\frac{\mathrm{d}m}{\mathrm{d}r} = 4\pi r^2 \rho(r) \tag{2}$$

The governing equation of state must be able to relate the radiation pressure to the density of the star. For this, we need to express the relation between pressure and radius (shown in Eq. (1)) as an expression related to the density of the gas. The white dwarf's outward pressure is supported by a degenerate electron gas. This means the pressure is inherently independent of temperature and only dependent on density, $P \propto \rho^{\frac{5}{3}}$ in non-relativistic regimes and $P \propto \rho^{\frac{4}{3}}$ in extreme relativistic regimes. Due to this, we can model the degenerate electron gas with density using the chain rule:

$$\frac{\mathrm{d}P}{\mathrm{d}r} = \frac{\mathrm{d}P}{\mathrm{d}\rho}\frac{\mathrm{d}\rho}{\mathrm{d}r} \tag{3}$$

The derivative for pressure with respect to density is given by the following equation;

$$\frac{\mathrm{d}P}{\mathrm{d}\rho} = Y_e \frac{m_e c^2}{m_p}\gamma\left(\frac{\rho}{\rho_0}\right) \tag{4}$$

Where c represents the speed of light in a vacuum, Y_e is the electron fraction, which can be quantified as the ratio between electrons and nucleons ($Y_e = \frac{\text{electrons}}{\text{nucleons}}$). The constants, m_e and m_p are the masses of electrons and protons, respectively. As well as this, ρ_0 is the fundamental density scale defined as $\rho_0 = \frac{m_p m_e^3 c^3}{3\pi^2 \hbar^3 Y_e}$ kgm^{-3} where $\hbar$ is the reduced Planck's constant. Finally, the $\gamma(y)$ function is a function that is used to capture the change between non-relativistic and extreme relativistic regimes and is defined as follows:

$$\gamma(y) = \frac{y^{\frac{2}{3}}}{3\left(1 + y^{\frac{2}{3}}\right)^{\frac{1}{2}}} \tag{5}$$

The purpose of the gamma function is to allow the model to predict the mass-radius relationship effectively as we move further from a non-relativistic regime (at lower masses) to a more extreme relativistic regime (at higher masses that approach the Chandrasekhar limit).

With this we now obtain 2 expressions for the equation of state of the white dwarf that can be written as a system of differential equations as shown below:

$$\frac{\mathrm{d}\rho}{\mathrm{d}r} Y_e \frac{m_e c^2}{m_p}\gamma\left(\frac{\rho}{\rho_0}\right) = -\frac{Gm(r)}{r^2}\rho(r) \tag{6}$$

[2]

III. NUMERICAL METHODS

From the differential equation Eq. (6), we now have to implement techniques derived for numerical approximation to solutions for this system, as an analytical solution is not easily defined. In this paper, C++ is used for numerical approximation of the equation's solutions, while Python is used for data visualisation.

This paper implements the Runge-Kutta 4th-order integration method in C++. The algorithm is arguably one of the most stable and widely praised numerical methods when solving differential equations. The choice for the use of Runge Kutta 4 is mainly due to its balance between accuracy to the system and its reasonable computational complexity. The algorithm focuses on solutions to Initial Value Problems (IVP), taking weighted averages of the slopes evaluated at the initial, final and midpoints between two points to evaluate a final approximation for the function at a different time. For a function $f(x_i, t_i)$, the Runge-Kutta approximation states that 4 slopes need to be calculated to receive a value for $f(x_{i+1}, t_i + \mathrm{d}t)$:

$$k_1 = f(x_i, t_i)$$

$$k_2 = f\left(x_i + k_1\frac{\mathrm{d}t}{2}, t_i + \frac{\mathrm{d}t}{2}\right)$$

$$k_3 = f\left(x_i + k_2\frac{\mathrm{d}t}{2}, t_i + \frac{\mathrm{d}t}{2}\right)$$

$$k_4 = f(x_i + k_3(\mathrm{d}t), t_i + (\mathrm{d}t))$$

By which we can approximate a value for x_{i+1} as:

$$x_{i+1} = x_i + \frac{\mathrm{d}t}{6}(k_1 + 2k_2 + 2k_3 + k_4) \tag{7}$$

[3]

While the benefits of the Runge Kutta 4 methods are plenty, they cannot overcome the limitations that computers bear when dealing with floating point arithmetic, which is often known as floating point errors. In this experiment we deal with densities of order 10^6 kg m^{-3}, solar masses that are of orders of magnitude 10^{30} kg while also dealing with proton and electron masses which are of order 10^{-27} kg and 10^{-31} kg respectively, the

sheer range of the orders of magnitudes involve make it so that using Runge Kutta Integration on the system of equations will result in floating point errors dominating and issues with selecting correct step sizes. As such, to avoid this issue, we have to utilise dimensionless variables to form a system of non-dimensionalised differential equations. We will be utilising a set of 3 dimensionless variables that will be substituted into Eq. (2) and Eq. (6). The substitutions involved are as follows:

$$r = R_0 \tilde{r} \tag{8}$$

where $R_0 = \sqrt{\frac{3Y_e m_e c^2}{4\pi G m_p \rho_0}}$ and $\tilde{r}$ is the dimensionless radius.

$$m = M_0 \tilde{m} \tag{9}$$

where $M_0 = \frac{4\pi\rho_0 R_0^3}{3}$ which is obtained due to the spherically symmetric shape of the star. While $\tilde{m}$ is the dimensionless mass.

$$\rho = \rho_0 y \tag{10}$$

where $\rho_0 = \frac{m_p m_e^3 c^3}{3\pi^2 \hbar^3 Y_e}$ kgm^{-3} and y is defined as the ratio between the central density and fundamental density scale $y = \frac{\rho}{\rho_0}$.

To obtain the non-dimensionalised equations (and the value for R_0), we must substitute Eq. (8) and Eq. (1)(0) (alongside the results $\mathrm{d}\rho = \rho_0\,\mathrm{d}y$ and $\mathrm{d}r = R_0\,\mathrm{d}\tilde{r}$) into Eq. (6) and arrange so that R_0 is a value that cancels all constants and scale factors leading to:

$$\frac{\mathrm{d}y}{\mathrm{d}\tilde{r}} = -\frac{my}{\tilde{r}^2 y} \tag{11}$$

As well as this, we need to substitute our expression for dimensionless mass using Eq. (9), Eq. (1)(0) and Eq. (8) into Eq. (2) to obtain a non-dimensionalised mass differential equation (which is calculated by substituting M_0 definition alongside the scaled equations. This results in the following equation:

$$\frac{\mathrm{d}\tilde{m}}{\mathrm{d}\tilde{r}} = 3\tilde{r}^2 y \tag{12}$$

We then assign an initial star structure with 0 dimensionless mass, a very small dimensionless radius (10^-6) and a dimensionless density varying from 10^{-2} to 10^6 and use Runge Kutta integration across space in small steps $\mathrm{d}\tilde{r} = 10^{-4}$ until dimensionless density approaches zero in the system of equations (signifying the star's surface).

Finally, using our definitions for R_0 and M_0 alongside the relevant electron fraction Y_e for the composition of the star (for Carbon-12 $Y_e = 0.5$ and for Iron-56 $Y_e = 0.464$) we obtain the relevant mass and radius for each central density (as our range of dimensionless densities represents a ratio between central density and the fundamental density scale).

This data is then plotted within a Python script, where the highest density ratios allow us to extrapolate values for the Chandrasekhar limit for the white dwarfs with Iron and Carbon compositions, respectively.

IV. RESULTS AND DISCUSSION

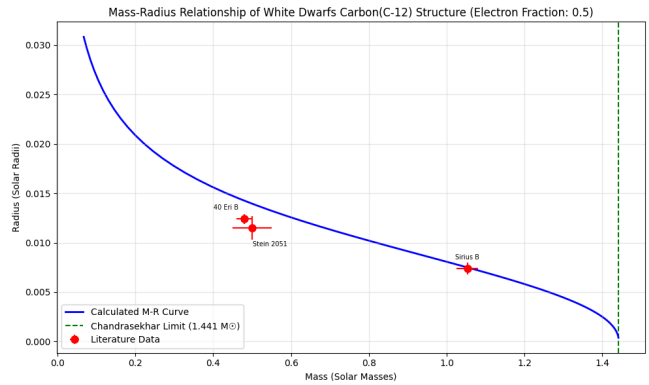

FIG. 1: Mass-Radius relationship between white dwarfs composed of Carbon 12.

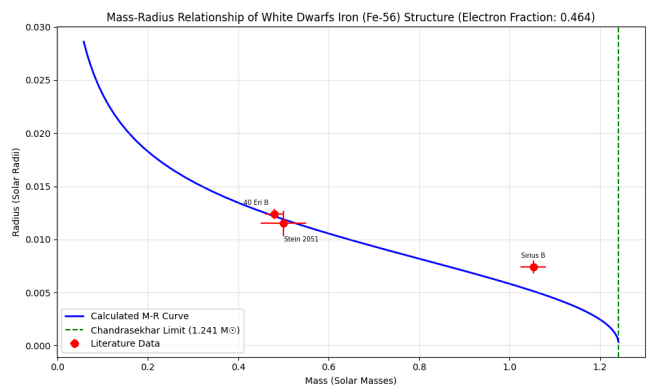

FIG. 2: Mass-Radius relationship between white dwarfs composed of Iron-56.

Figure 1 represents the mass-radius (in terms of solar masses $M_\odot = 1.98 \times 10^{30}$ kg and solar radii $R_\odot = 6.95 \times 10^8$ m) relationship obtained for Carbon Rich compositions (with an electron fraction $Y_e = 0.5$ and an estimate for the Chandrasekhar limit near 1.44 $M_\odot$). Figure 2 shows the mass radius relationship obtained for more iron-abundant white dwarfs (with an electron fraction $Y_e = 0.5$ and an estimate for the Chandrasekhar limit near 1.24 $M_\odot$), respectively.

The graphs both indicate an inverse relationship between mass and radius, physically (and as shown by Eq. (1) this is expected. The reason for this relation is due to the fact that stellar radius decreases when only gravitational forces dominate electron degeneracy pressure, and a new equilibrium state where electrons are at higher energy states (resulting in an increasing electron degeneracy pressure). As such, both electron degeneracy pressure and mass of the stellar remnant are inversely proportional to the radius.

TABLE I.Mass-Radius data for 3 known white dwarfs

	Mass	**Radius**
Sirius B	1.053 ± 0.028	0.0074 ± 0.0006
40 Eri B	0.48 ± 0.02	0.0124 ± 0.0005
Stein 2051	0.50 ± 0.05	0.0115 ± 0.0012

[2]

To validate the numerical model and be able to approximate compositions of known white dwarfs, we are given in Table 1. We see that 40 Eri B and Stein 2051b align (within the range of error) with iron-rich core compositions. Sirius B, however, tends to align (within range of error) with the mass radius relationship for cores composed of Carbon.

The Chandrasekhar limit can be approximated by the following formula:

$$M_{\text{ch}} \approx 5.77(Y_e)^2 M_\odot \tag{13}$$

Lower mass white dwarfs align remarkably well with iron-rich compositions, which is further justified by the Chandrasekhar limit for Iron Core data, which can be approximated by Eq. (1) (3) to $M_{\text{ch}} \approx 1.24 M_\odot$. Meanwhile, higher mass white dwarfs align well with carbon-rich compositions for which we obtain $M_{\text{ch}} \approx 1.44 M_\odot$. This model does have significant limitations in aspects due to underlying assumptions, especially regarding a lack of relativistic corrections, and not accounting for the rotations, as well as the ideal gas composition being assumed.

V. CONCLUSION

In this study we modelled successfully the structure of white dwarfs using Runge-Kutta 4th order integration for a system of unsolveable differential equations using C++ and plotted the resultant inverse mass-radius relationship for Iron composed white dwarfs and carbon-composed white dwarfs through Python. This was then compared against given masses for known white dwarfs (with ranges of error), which implied that 40 Eri B and Stein 2051b are white dwarfs with an Iron composition, and Sirius B is a white dwarf with a Carbon composition. We also obtained estimated values for the Chandrasekhar limit beyond which the hydrostatic equilibrium between electron degeneracy pressure (as a consequence of Pauli's Exclusion principle) and gravitational forces (due to the mass of the stellar remnant) collapses. For iron-rich white dwarfs, this value is around 1.24 $M_\odot$ which we believe is an accurate approximation as it agrees with the theoretical approximation. Similarly, carbon-rich white dwarfs lead to an approximate Chandrasekhar limit of $1.44 M_\odot$, which is one chain an accurate approximation due to its agreement with the theoretical approximation. Despite the apparent excellence of the model, it fails to account for relativistic effects and rotational properties of white dwarfs, which could be an area of improvement to consider when designing more accurate simulations of white dwarf structures. We could also present improvements by using adaptive stepping in the Runge-Kutta 4th order implementation alongside considering higher order terms to help approximate surfaces to greater detail.

VI. ANNEX

A. Random Walks

Random walks are a mathematical modelling technique by which we can describe paths consisting of random steps in a mathematical space.

Historically, the use of random walks has always been focused towards infinite-sized, d-dimensional spaces, and classical theory focuses on how differences occur in the way the random walk returns. When we look towards where d = 1 and d = 2, via $Pòlya's$ theorem, we find that (with probability 1) we return to the starting point infinitely often.[4] However, when we look towards higher-dimensional spaces ($d \geq 3$), we see that this behaviour tends towards a finite number of returns. [5]

That being said, the idea to model random walks as traversals on graphs, more specifically, traversals on finite graphs, has gained attention and the uses in which are focused around quantitative measures such as the time taken to traverse all nodes (known as cover time), among other important properties. [6]

One of the most famous uses of these is the idea of electrical resistance modelling, where we can model the effective resistance between 2 nodes by the difficulty of a random walk between them, where a higher resistance corresponds to a lower probability of traversal between the nodes. [5]

The notion of treating lower traversal as akin to an effective resistance allows for a plethora of use cases. One of the most important use cases of the idea of cover time and effective resistance is the theory of communication networks, in which effective resistance can be used to judge fault tolerance and congestion between nodes and throughout entire routes. This is essential in building computational communication networks and protocols that are at the heart of modern communications. It is a foundational idea for some machine learning techniques, such as spectral clustering, as well.[7]

The idea of random walks and resistance distance (the idea of measuring all possible paths to obtain an idea of how far 2 nodes are through modelling a graph as a circuit) can be applied in areas of biology and geology as well. By modelling rough terrains as areas of high resistance and smoother terrains as areas of lower resistance, we predict how connected 2 populations are across terrains. The application of this theory extends further, as we can model how genetic data transfer across landscapes and provide a more accurate description of genetic diversity across landscapes than with earlier Euclidean distance or simpler weighted-path theories. [8]

One of the most important uses of random walks within our lives today is in Recommender systems. By once again taking a graphical approach towards having nodes that are similar or often closer to each other within the graph, we can infer relationships based on distance and connectivity. These random

walks form into 3 main categories. Global Recommender Systems provide a single ranking scheme for every node in the system and are often not useful for individual customisation. PageRank (a system made for optimising search engines) is the most famous example of a Global Recommender System. Random Walks with Restarts (RWR) is a more customisable approach to recommender systems, as it encodes a mathematical preference towards certain outcomes after restarts. One of the most important use cases for this type of Recommender System is in Location-Based Social Networks (LSBNs) like Snapchat's famous Snap Maps. [9]

Random walks are also important in discussing effective optical depth (the point at which photons are last scattered before escaping). Random walks are used to model regions where photon scattering is dominant, as photons are described to take a random walk before escaping, making it so that the effective distance they travel is far greater than the actual thickness they must pass through to escape. We can take advantage of this fact to be able to predict supernovae hours before they happen. This can be done by using neutrino detectors instead (as neutrinos are weakly interacting while also travelling very close to the speed of light), while radiation from the supernova is often scattered heavily and has to travel far longer distances to reach Earth; this is not the case for neutrinos. They often do not scatter, so detecting neutrinos can help detect supernovae before we even visibly see the effects. [10,11]

Random walks have broad areas of use within game design as well. Random walks take up an important space within open-world games, as most scene-changing and decision-making algorithms for such games rely on random walks as well as interactions with non-playable characters. Games which are more focused on decision-making, like visual novels, often rely on random walks to map conversations and routes that characters will take as well. These uses for random walks have now become the foundation of modern game design in many industries. [12]

B. Monte Carlo Methods

Monte Carlo Methods are a series of numerical analysis techniques by which we can use random sampling processes in complex mathematical problems which are either hard or impossible to analytically solve. With it, we are able to statistically predict likely outcomes within the system. [13]

Monte Carlo Methods can be used in tandem with Markov Chains. Markov Chains are another mathematical method that dictates a sequence of possible events, where each state can only evolve into the next state based on a transition probability of the current state. Markov Chain Monte Carlo methods (MCMC methods) hold grave importance in statistical physics. The Metropolis-Hastening Algorithm is one such method whereby you have a distribution which you wish to sample from, that is, a Markov chain with a proposed distribution by which you transition to a new state, then calculate the acceptance probability between your target and proposed distribution iteratively. This method was first used in statistical physics when dealing with thermal fluctuations in a collection of atoms modelled as 2-dimensional circular particles, in which all states of a collection of these atoms were computed, where random changes to states after steps would only be accepted firmly if a lower energy configuration is reached with a change to accept a higher energy configuration that depends on a probability proportional to $e^{-\frac{\Delta E}{\mathrm{d}t}}$. [13]

The success story of AlphaGo is one of the greatest tributes to the usage of Monte Carlo Methods in Deep Learning. AlphaGo is a model that uses Monte Carlo Tree Searches, which is a decision-making algorithm that focuses on a four-phase approach (selection, expansion, simulation and backpropagation) to generate lists of possible states which provide the highest reward. AlphaGo became a revolutionary paper as it showed that training Deep Learning models could help the models overcome complex and evolving situations even better than the greatest professionals. The paper would go on to be a benchmark for both Go and chess engines, including the development of AlphaZero, which eventually learned to play chess against and was able to best Stockfish decisively within hours, ushering in a new era for game development. [14]

Lattice Quantum Chromodynamics (LQCD) is another major area in which Markov Chain Monte Carlo methods are essential. Lattice QCD is a computational attempt at trying to approximate solutions numerically to a low-energy non-perturbative regime where gluon-gluon interactions often prevent an analytical solution from being reached for a system due to the coupling of the strong nuclear force. Hadron masses (such as proton and neutron masses) are often far heavier than the sum of the masses of their quark constituents. This discrepancy itself was solved with the help of the techniques of the Hamiltonian Monte Carlo method (HMC), which is a highly efficient Monte Carlo variant adept at dealing with highly complex problems. From the use of the HMC method across multiple different lattices, it was observed that 95% of the mass of hadrons comes from gluon-gluon interactions. [15]

Finance is an industry that has shown that Monte Carlo Methods display more than just the ability for frontier-defining theoretical physics. Options are derivative contracts that allow the right but not the obligation to buy or sell assets within a timeframe. Options pricing is a very important part of financial markets. In cases where analytical solutions, such as the Black-Scholes equation, cannot be applied easily, we have to use Monte Carlo methods in tandem with Geometric Brownian Motion (a method that uses volatility, drift and statistical shocks applied to a stock price to show how asset pricing evolves) to help introduce more randomness on different days, from which we can simulate and see how trends suggest that stock prices will change generating a plethora of potential paths of which most likely paths are often preferred and an average is projected onwards. [16]

The uses of Monte Carlo methods are not limited to physics alone. Gibbs Sampling, a Markov Chain Monte Carlo method

that focuses on sampling from a collection of 1-Dimensional simple distributions to emulate samples from a more complex multidimensional distribution, is often used in protein folding algorithms. The Rosetta algorithm uses Gibbs sampling to help sample small protein chains and be able to construct predictions for larger protein structures from these chains, and is widely regarded as one of the greatest algorithms for protein structure prediction. [17]

Photorealistic rendering is a field in which we have seen remarkable advancements towards making fields such as Computer Generated Imagery (CGI). One of the main driving forces behind the realistic textures of modern CGI is related to Monte Carlo methods as well. Monte Carlo Ray Tracing is a technique by which objects are modelled with thousands of rays of light that interact with the object to form multiple images, and performing Monte Carlo Integration (estimating average values of functions and integrating over the function). This can result in reductions in motion blur, focusing issues, creation of soft shadows, all of which are issues that standard ray tracing methods could not resolve. This method also allows for a greater generation of depth in satellite imaging. [18]

C. Random Numbers

Random numbers are often the foundation for most computational methods. True randomness, however, is quite hard to achieve within controlled environments. This is especially true for digital machines such as computers. As such, most systems rely on pseudo-random number generation for stochastic (dependent on random behaviour) methods. Alternatively, one can use some form of random process (such as atomic decay) as a determining factor for these random number generators. [19]

One of the proposed solutions towards random number generation lies within the field of Quantum Physics. Quantum Random Number Generators (QRNGs) are an area of research by which we can use quantum processes to generate device-independent random numbers while having a framework by which we can actively state that there are no hidden variables that directly influence the randomness of a quantum system, as shown by John Bell in 1964. [20] However, such systems are not without drawbacks, as there is a fundamental difficulty with handling sources that act on a quantum scale, preserving output and limiting thermal noise, which are areas in which research is being dedicated. [21]

Cryptography is a key area in which high-quality pseudo-random number generation is essential. The primary goal of their use is to prevent key reversal and to have truly secure keys, IDs, and digital signatures as well. We require random numbers to prevent attackers from exploiting the keys. The distinction between what one would consider pseudo-random numbers in many computational methods is not nearly good enough at preventing attackers from using methods such as iterative guessing or chosen input attacks, so random number generators like the MT19937 (a default pseudo-random number generator in most applications and programming languages) have predictable states (624 32-bit registers and attackers can observe each to figure out the internal state). Cryptographically Secure Pseudo-Random Number Generators (CSPRNGs) are necessary. An example of this would be Yarrow, which takes streams of high entropy, unpredictable data and stores it (such as keystrokes, mouse clicks and others), then splits into a higher output lower entropy pool and a lower output higher entropy pool. It constantly changes seeds once it gathers enough data to reduce predictability and timing of windows. Finally, it outputs through block ciphers. [22]

The idea of perceived randomness is an important topic when addressing randomness in computational sciences, especially in computational biology. Most influential biologists of the late 20th century believed that the smaller a system is, the less likely it is to be truly random, and that complexity in completing similar tasks would be the main factor that determines true randomness. This belief, which had been supported by many biologists throughout the 1960s and 1970s, is what inspired many to talk about the idea of perceived randomness. Humans will subconsciously try to find patterns in complex statistical scenarios, and when it is proving to be extremely hard to do so, we believe the process is random when in reality it could either be random or chaotic and complex, but predictable if the right tools are provided (this is a large area of concern within True Random Number Generation). Behavioural Game Theory is a branch of Game Theory which focuses on the trust that we put towards the randomness of games and if that is exploitable to determine winning outcomes. [23,24]

Simulating randomness in weather and generating a model that can be used to monitor practices for agricultural practices across seasons, amidst monitoring and simulating anomalies in weather data before forecasting. Markov chains and random number generation are used with constraints regarding position and season to generate a model that can predict aspects like precipitation. While multidimensional stochastic modelling techniques can be used to generate data on solar radiation, minimum and maximum temperatures and wind data. The benefits of such a model for weather generation stem from day-to-day life for forecasters and are especially important within agriculture, where being able to predict how crops will perform over seasons is essential to help farmers with their produce. [25]

Simulation of complex environments depends heavily on random sampling methods. Stochastic processes often govern the process by which we sample extensively. However, processes can be limited by the computing resources they require and not all processes can be run with extensive simulations before use. For processes where only a few simulations can be run, techniques such as Latin Hypercube Sampling (LHS) are used, where we use random numbers to generate points across a multidimensional space of variables, ensuring even sampling across multiple variables. This technique focuses on a stratified coverage rather than a simple random sample's more clustered approach. It holds extreme importance in modelling complex real-world systems (especially within structural engineering) to

an extensive degree without the use of thousands of simulations. [26]

Logistics is an industry focused on complex transport chains intertwined with centralised sorting, making it prone to failures. One can use random number sampling alongside Markov Chains to simulate failures and breakdowns within complex distribution systems, and their effects on the overall system are simulated using Monte Carlo Methods. Discrete Event Simulation is an example of this simulation system. Random bottlenecks (such as weather or breakdowns) are simulated with random number generation and Markov Chains for the effects within one area of the distribution system. To simulate its effects on the whole system, simulators run Monte Carlo simulations on the entire system to predict the most likely effects and worst-case scenarios to prepare for. This helps prevent system collapse even in unprecedented circumstances. [27]

CODE AVAILABILITY

The full implementation for this project is The full implementation used in this project is available in the following Public GitHub repository:

https://github.com/Krishk1223/White-Dwarfs

The repository contains:

- C++ source code implementing the RK4 integration method
- Python scripts used for data generation and plotting
- Instructions for compilation and execution
- A feature by which you can simulate planetary orbits of planets the size of Earth around a white dwarf of a mass of your choice (this is still currently being refined) using Raylib.

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